



Platform Smart Farming

Capstone Project Documentation

Compiled by:

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Technology Stac	MERN (MongoDB, Express, React, Node.js)
Architecture	MVC (Model-View-Controller)
Authentication	JWT + Bcrypt
External API	OpenWeatherMap API
Theme	Sustainability & Green Tech

Sustainability & Green Tech | Capstone Project

BAB 1 — Introduction & Problem Statement

1.1 Background

Indonesia has approximately 33 million farmers, 60% of whom are smallholders with land under 0.5 hectares. Although agriculture contributes 13% of national GDP and absorbs 29% of the workforce, most smallholders still face structural challenges that hinder their productivity and welfare. Traditional farming methods heavily rely on manual recording and experience-based decisions. Consequently, farmers often face: harvest losses due to undetected plant diseases early on, waste of fertilizer and water due to lack of data-driven guidance, ignorance of market prices which puts them at a disadvantage to middlemen, and isolation from modern agricultural knowledge.

1.2 Problem Statement

Farmer's Problem	AgriTech Solution
No real-time crop monitoring system	Crop Monitoring with stage & health tracking, automatic alerts
Unaware of correct fertilizer dosage and irrigation schedule	Recommendation Engine based on crop growth phase
No access to accurate weather information	OpenWeatherMap API integration — auto-detect & city search
Unaware of market prices, often disadvantaged by middlemen	Market Price Module — 8 commodities with trend graphs
Isolated from other farmers and agricultural experts	Community Forum + Integrated Expert Advice
Difficulty buying agricultural inputs at competitive prices	Resource Purchase Module — buy fertilizer, seeds, tools

1.3 Tujuan & Ruang Lingkup

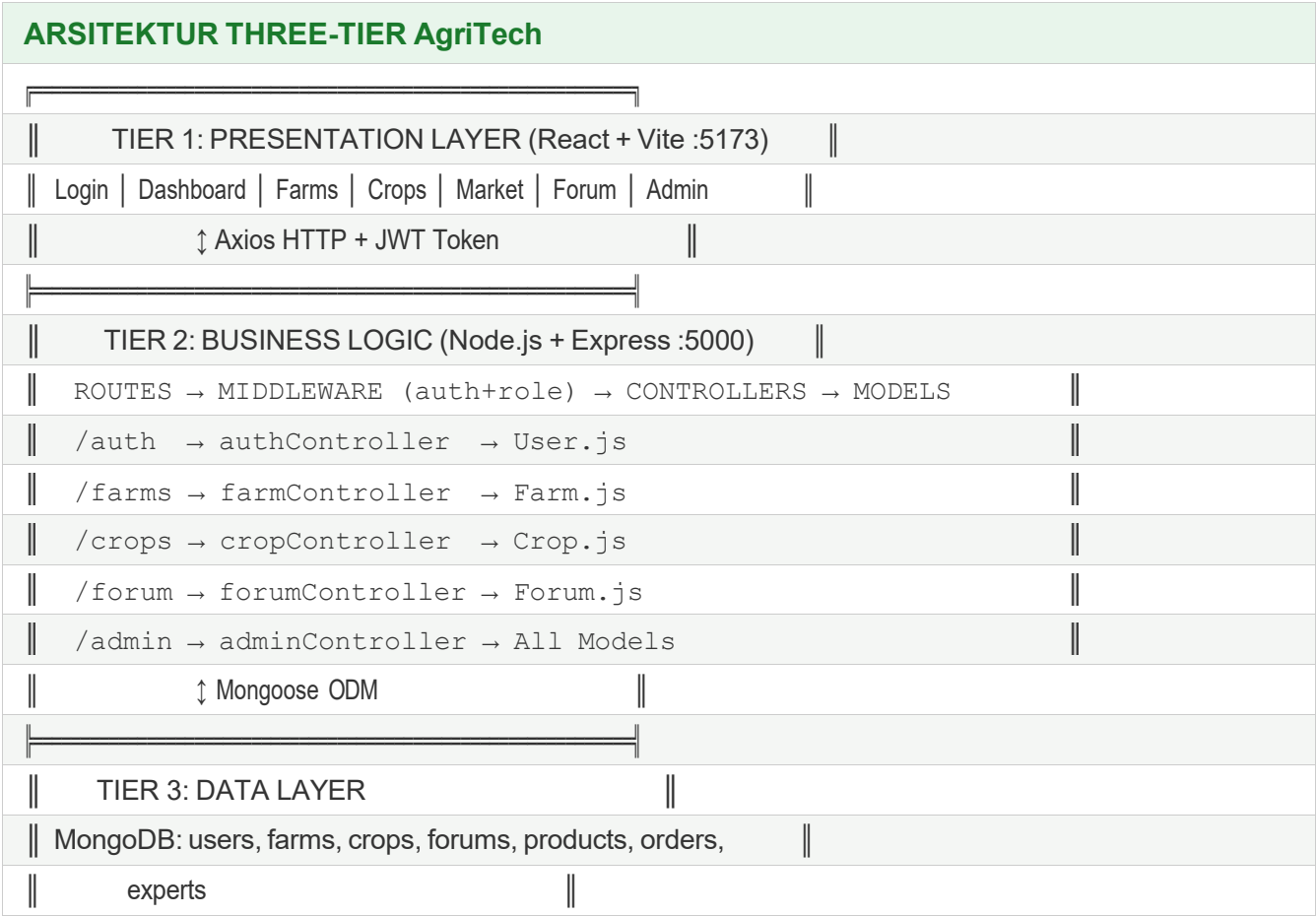
- Build an integrated digital platform that consolidates all farmer information needs
- Implement a real-time crop monitoring system with automatic threshold-based alerts
- Build a secure authentication system with role-based access control (RBAC)
- Integrate external data (weather, market prices) to support decision-making
- Facilitate a farmer community and access to agricultural expert consultations
- Provide data-driven recommendations for fertilizer and irrigation optimization according to planting phase

BAB 2 — System Architecture

2.1 Perbandingan MERN vs Alternatif

Aspek	MERN Stack (Dipilih)	LAMP Stack (Tidak Dipilih)
Language	JavaScript across all layers	PHP backend + JS frontend
Database	MongoDB — NoSQL, flexible schema	MySQL — SQL, rigid schema
Performance	Non-blocking I/O, suitable for REST API	Synchronous, less efficient
Scalability	Horizontal scaling easy	Vertical scaling more common
Real-time	Socket.io easy to integrate	Requires complex configuration
Decision	☑ for modern Agri-Tech	✗ Not Selected

2.2 Diagram Arsitektur Sistem



External: OpenWeatherMap API
=====

2.3 Implementasi MVC

Layer MVC	File	Responsibility
MODEL	models/User.js models/Farm.js models/Crop.js models/Forum.js models/Product.js models/Order.js models/Expert.js	Defines data structure, field validation, and relationships between collections in MongoDB using Mongoose Schema
CONTROLLER	controllers/authController.js controllers/farmController.js controllers/cropController.js controllers/dashboardController.js controllers/weatherController.js controllers/forumController.js controllers/productController.js controllers/expertController.js controllers/adminController.js	Handles business logic: receives requests from routes, processes data via models, returns JSON response to client
VIEW (Frontend)	pages/Login.jsx pages/Dashboard.jsx pages/Farm.jsx pages/Crop.jsx pages/Recommendations.jsx pages/MarketPrice.jsx pages/Market.jsx pages/Forum.jsx pages/ExpertAdvice.jsx pages/Admin.jsx	React components that display data to users and handle UI interactions. Completely separated from backend

2.4 Architecture Scalability

Aspect	Implementation
Stateless Backend	JWT token verified on each request — no session on server. Multiple instances can run in parallel.
MongoDB Scaling	Supports sharding and replica sets for data distribution and redundancy.
Modular Architecture	Each feature encapsulated in a separate module — adding new features does not affect other modules.
Connection Pooling	Mongoose manages connection pool to MongoDB — connections are reused, not recreated per request.
Environment Config	All configurations in `.env` — easy deployment to various environments (dev, staging, production).

BAB 3 — Frontend Development

3.1 React Component Structure

Komponen / Halaman	React Hooks & State Management
Login.jsx	<code>`useState`</code> : tab, email, password, loading; <code>`useState`</code> : redirect to <code>`/dashboard`</code> after login; <code>`useState`</code> : not used (no side effects)
Dashboard.jsx	<code>`useState`</code> : data, weather, cityInput, loading; <code>`useState`</code> : fetch <code>`/dashboard`</code> + <code>`/weather`</code> on mount; <code>`useState`</code> : for quick links
Farm.jsx	<code>`useState`</code> : farms[], name, location, size, editingId, loading; <code>`useState`</code> : <code>`fetchFarms()`</code> on component mount; API calls: GET, POST, PUT, DELETE <code>`/farms`</code>
Crop.jsx	<code>`useState`</code> : crops[], farms[], name, farmId, stage, healthStatus, loading; <code>`useState`</code> : <code>`loadFarms()`</code> + <code>`fetchCrops(farmId)`</code> when farmId changes; Derived state: <code>`sickCrops = crops.filter(c => c.healthStatus === 'sick')`</code>
Navbar.jsx	<code>`useLocation`</code> : detect active route for menu highlighting; <code>`useState`</code> : redirect to <code>`/`</code> on logout; <code>`localStorage`</code> : read user data to display name

3.2 Responsive Design

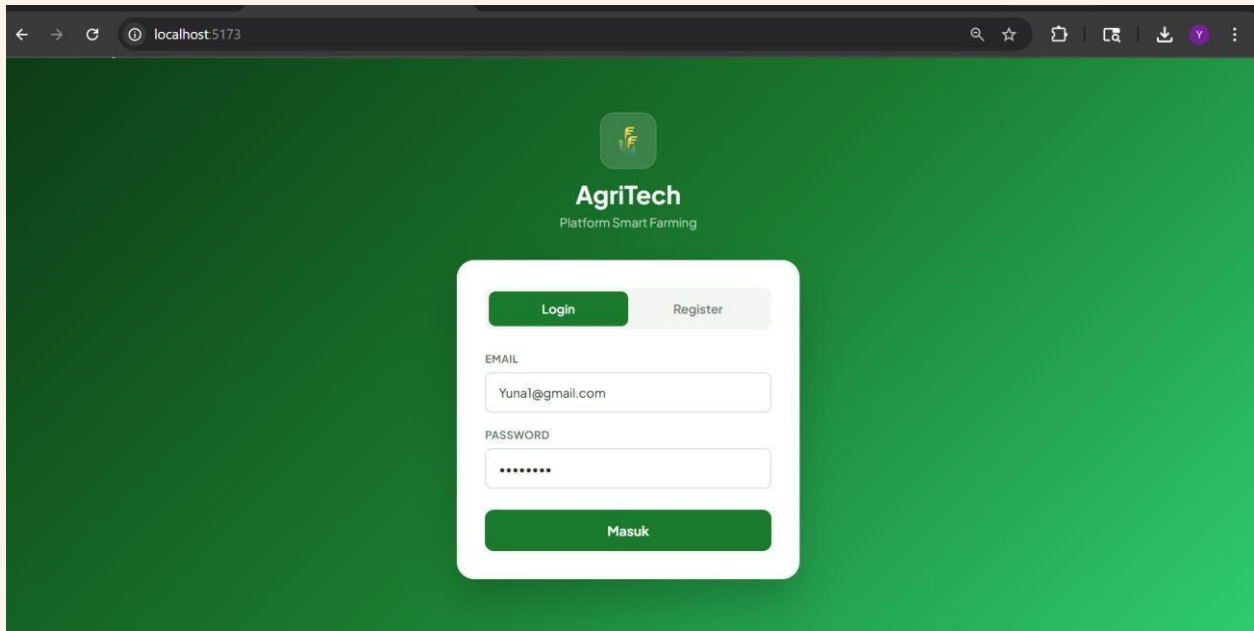
AgriTech uses Bootstrap 5 grid system to ensure responsive display across various screen sizes:

- ``col-md-3 col-6`` — Stat cards: 4 columns on desktop, 2 columns on mobile
- ``col-lg-5, col-lg-4, col-lg-3`` — Dashboard layout: 3 panels side-by-side on desktop
- ``col-md-4 col-lg-3`` — Product grid: 4 columns on desktop, 3 on tablet, 1 on mobile
- ``table-responsive`` — Tables can be scrolled horizontally on small screens

3.3 Screenshot UI

➤ Login & Register Page

SCREENSHOT: Login Page



A screenshot of a web browser displaying the login page for AgriTech. The browser's address bar shows 'localhost:5173'. The page has a green background with the AgriTech logo (a green square with a yellow 'F' and a plant) and the text 'AgriTech Platform Smart Farming' centered at the top. Below this, there is a white login form. The form has two tabs: 'Login' (active) and 'Register'. It contains fields for 'EMAIL' (with the value 'Yuna@gmail.com') and 'PASSWORD' (with masked characters '*****'). A green button labeled 'Masuk' is at the bottom of the form.

localhost:5173

AgriTech
Platform Smart Farming

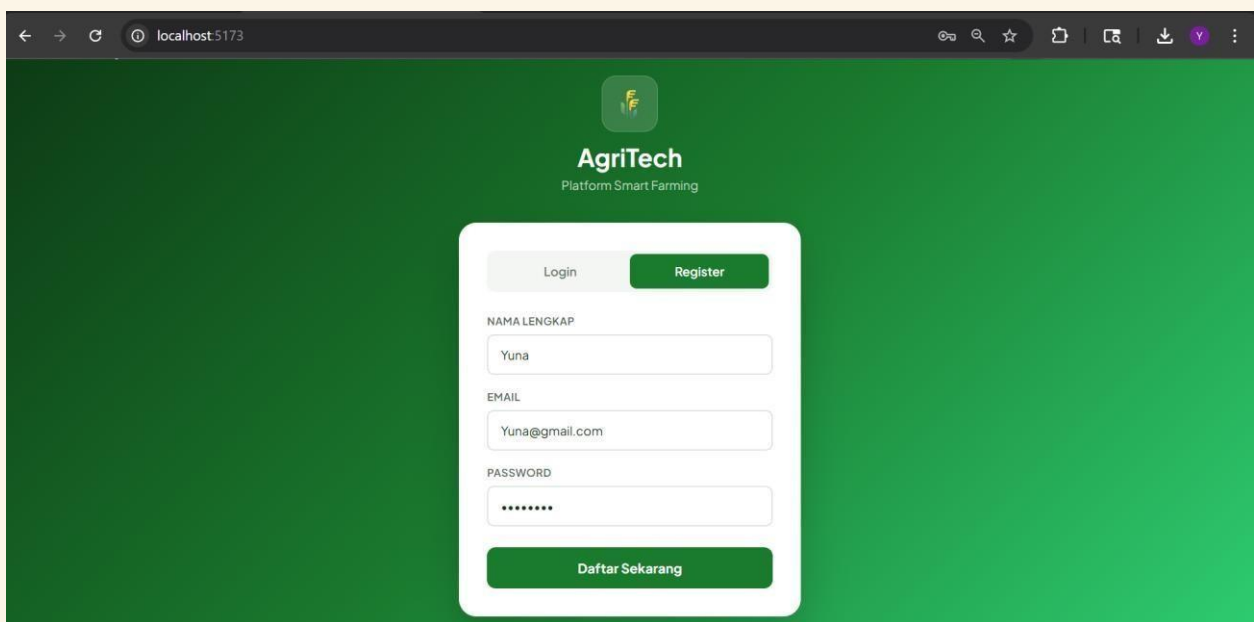
Login Register

EMAIL
Yuna@gmail.com

PASSWORD

Masuk

SCREENSHOT: Register Page



A screenshot of a web browser displaying the register page for AgriTech. The browser's address bar shows 'localhost:5173'. The page has a green background with the AgriTech logo (a green square with a yellow 'F' and a plant) and the text 'AgriTech Platform Smart Farming' centered at the top. Below this, there is a white registration form. The form has two tabs: 'Login' and 'Register' (active). It contains fields for 'NAMA LENGKAP' (with the value 'Yuna'), 'EMAIL' (with the value 'Yuna@gmail.com'), and 'PASSWORD' (with masked characters '*****'). A green button labeled 'Daftar Sekarang' is at the bottom of the form.

localhost:5173

AgriTech
Platform Smart Farming

Login Register

NAMA LENGKAP
Yuna

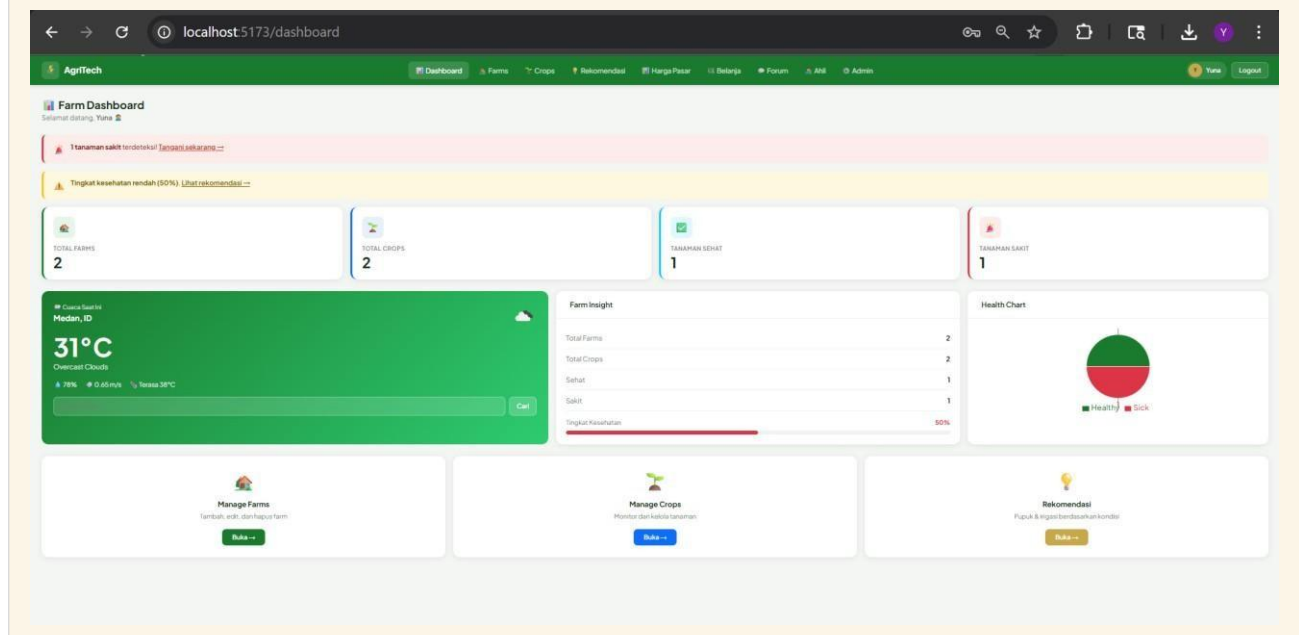
EMAIL
Yuna@gmail.com

PASSWORD

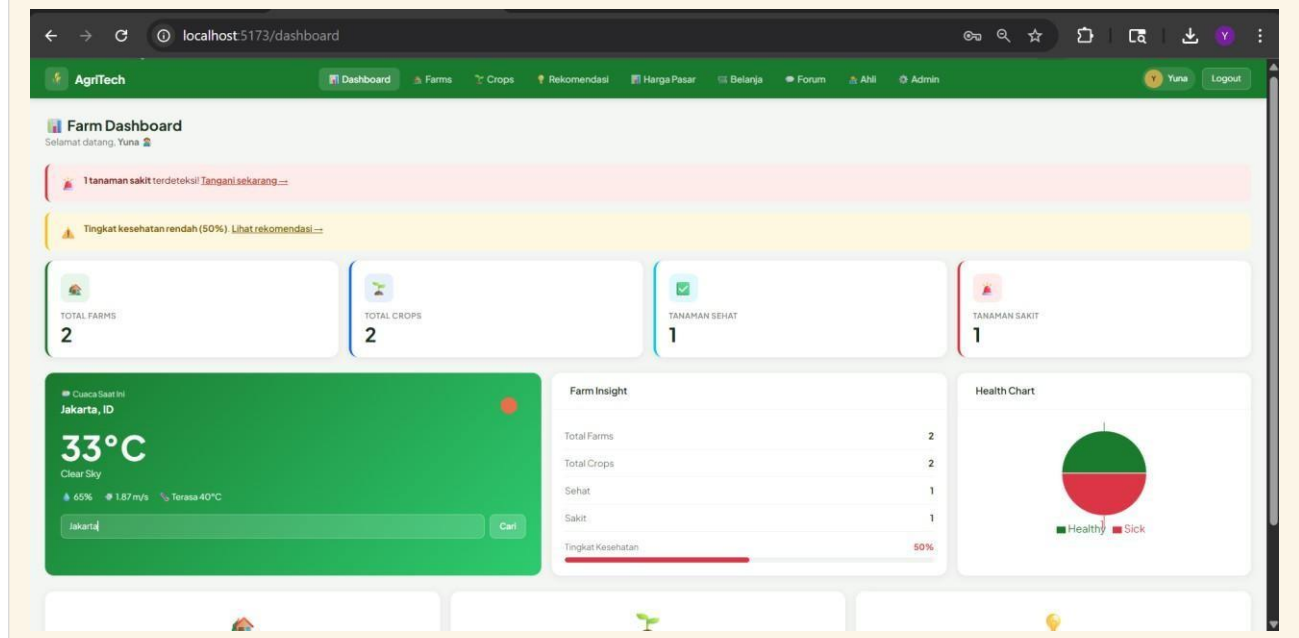
Daftar Sekarang

➤ Dashboard

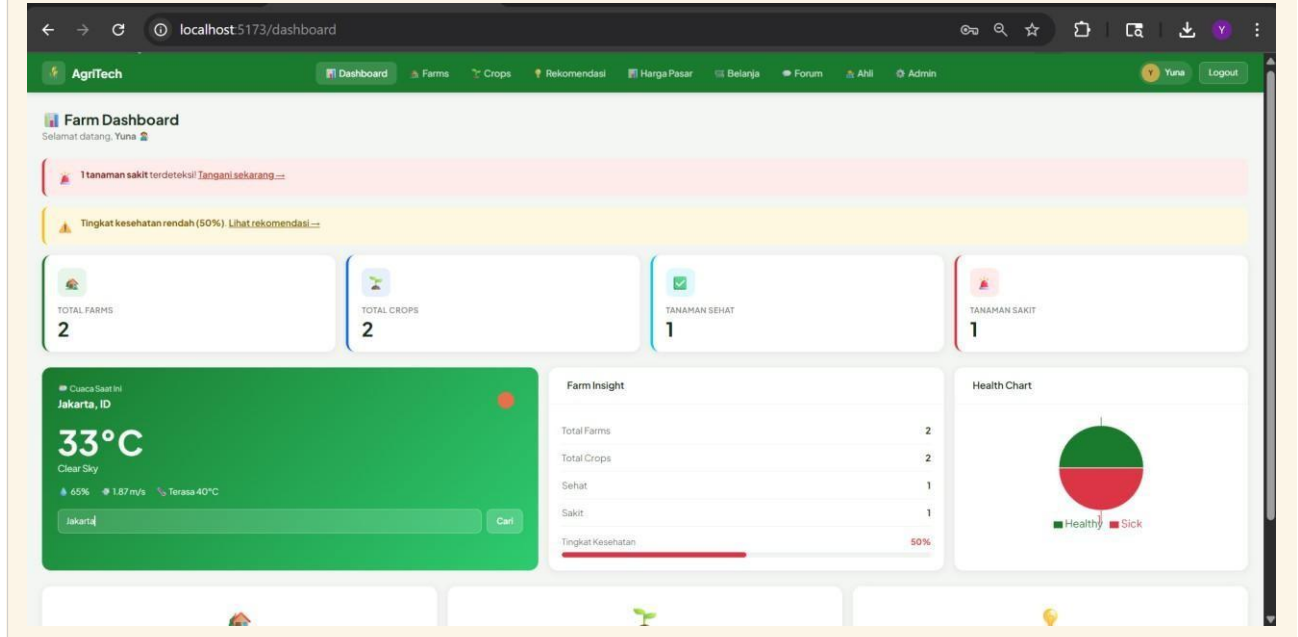
📸 SCREENSHOT: Dashboard — Stat Cards & Alert



📸 SCREENSHOT: Dashboard — Weather Widget



SCREENSHOT: Dashboard — Pie Chart



➤ Farm Management

SCREENSHOT: Farm Management — Daftar Farm

The screenshot shows the AgriTech Farm Management interface. The top navigation bar includes links to Dashboard, Farms, Crops, Rekomendasi, Harga Pasar, Belanja, Forum, Ahli, and Admin. The user is logged in as Yuna. The page displays the following information:

- Farm Management:** Kelola semua data farm Anda.
- Tambah Farm Baru:** Form to add a new farm with fields for NAMA FARM, LOKASI, and UKURAN (HEKTAR).
- Daftar Farm (3):** Table of existing farms.

#	NAMA FARM	LOKASI	UKURAN	DIBUAT	AKSI
1	Sawah Kentang	Deli Serdang	1 Ha	11/5/2026	Edit Hapus
2	Sawah Padi	Berastagi	3 Ha	6/5/2026	Edit Hapus
3	Sawah Jagung	Medan	1,5 Ha	6/5/2026	Edit Hapus

SCREENSHOT: Farm Management — Form Tambah

localhost:5173/farms

AgriTech Dashboard Farms Crops Rekomendasi Harga Pasar Belanja Forum Ahli Admin Yuna Logout

Farm Management

Kelola semua data farm Anda

+ Tambah Farm Baru

NAMA FARM: Sawah Tomat LOKASI: Berastagi UKURAN (HEKTAR): 13

+ Tambah Farm

Daftar Farm (3)

#	NAMA FARM	LOKASI	UKURAN	DIBUAT	AKSI
1	Sawah Kentang	Deli Serdang	1 Ha	11/5/2026	Edit Hapus
2	Sawah Padi	Berastagi	3 Ha	6/5/2026	Edit Hapus
3	Sawah Jagung	Medan	1.5 Ha	6/5/2026	Edit Hapus

➤ Crop Monitoring

SCREENSHOT: Crop Monitoring — Tabel Crop

localhost:5173/crops

AgriTech Dashboard Farms Crops Rekomendasi Harga Pasar Belanja Forum Ahli Admin Yuna Logout

Crop Management

Monitor dan kelola tanaman di farm Anda

1 tanaman sakit di farm ini: Padi
Segera lakukan penanganan atau cek rekomendasi.

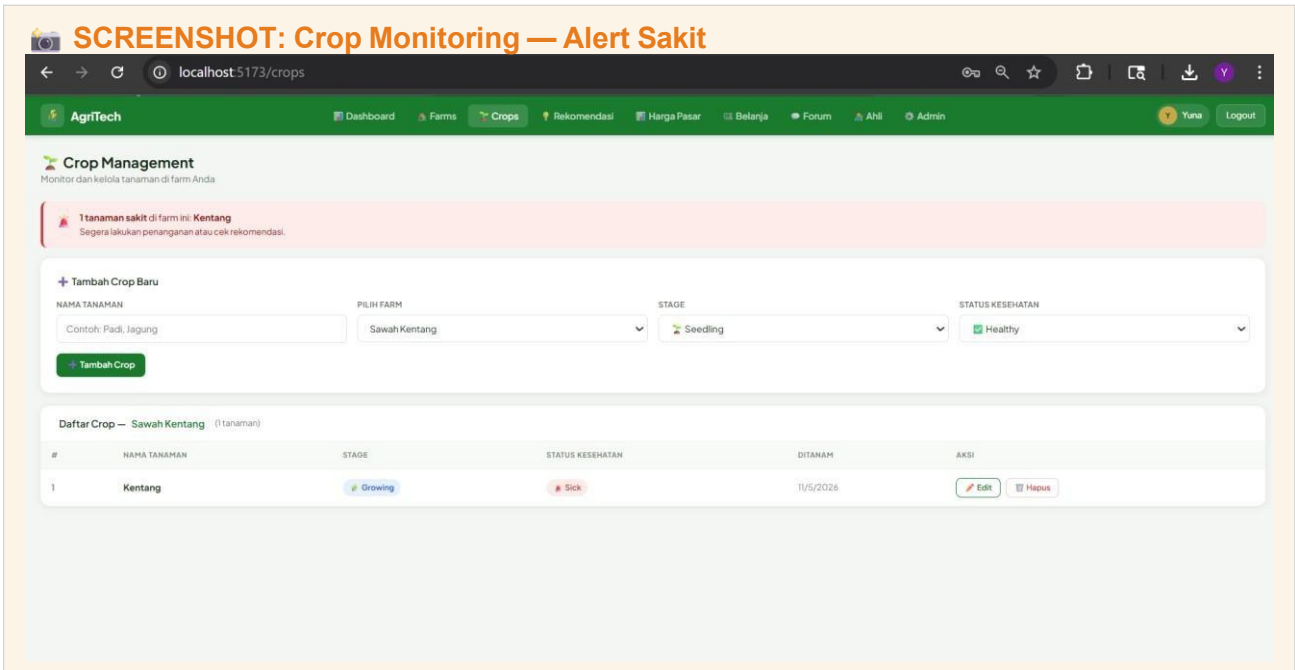
+ Tambah Crop Baru

NAMA TANAMAN: Contoh: Padi, jagung PILIH FARM: Sawah Padi STAGE: Seeding STATUS KESEHATAN: Healthy

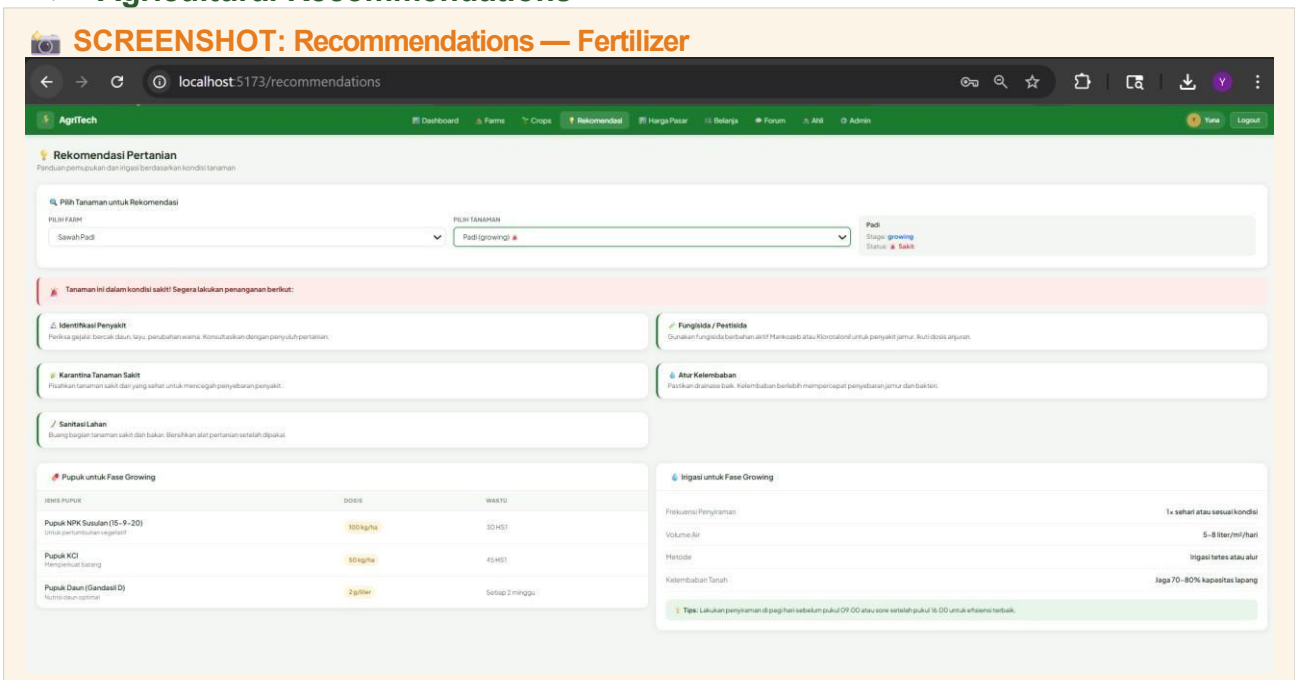
+ Tambah Crop

Daftar Crop — Sawah Padi (2 tanaman)

#	NAMA TANAMAN	STAGE	STATUS KESEHATAN	DITANAM	AKSI
1	Padi2	Seeding	Healthy	11/5/2026	Edit Hapus
2	Padi	Growing	Sick	6/5/2026	Edit Hapus



➤ Agricultural Recommendations



SCREENSHOT: Recommendation — Irrigation

AgriTech

DashboardFarmsCropsRekomendasiHarga PasarBelanjaForumAhlAdmin

YunaLogout

Status: Sakit

Tanaman ini dalam kondisi sakit! Segera lakukan penanganan berikut:

Identifikasi Penyakit

Periksa gejala: bercak daun, layu, perubahan warna. Konsultasikan dengan penyuluh pertanian.

Karantina Tanaman Sakit

Pisahkan tanaman sakit dari yang sehat untuk mencegah penyebaran penyakit.

Sanitasi Lahan

Buang bagian tanaman sakit dan bakar. Bersihkan alat pertanian setelah dipakai.

Fungisida / Pestisida

Gunakan fungisida berbahan aktif Mankozeb atau Klorotalonil untuk penyakit jamur. Ikuti dosis anjuran.

Atur Kelembaban

Pastikan drainase baik. Kelembaban berlebih mempercepat penyebaran jamur dan bakteri.

Pupuk untuk Fase Growing

JENIS PUPUK	DOSIS	WAKTU
Pupuk NPK Susulan (15-9-20) Untuk pertumbuhan vegetatif	100 kg/ha	30 HST
Pupuk KCI Memperkuat batang	50 kg/ha	45 HST
Pupuk Daun (Gandasil D) Nutrisi daun optimal	2 g/liter	Setiap 2 minggu

Irigasi untuk Fase Growing

Frekuensi Penyiraman

1x sehari atau sesuai kondisi

Volume Air

5-8 liter/m²/hari

Metode

Irigasi tetes atau alur

Kelembaban Tanah

Jaga 70-80% kapasitas lapang

Tips: Lakukan penyiraman di pagi hari sebelum pukul 09.00 atau sore setelah pukul 16.00 untuk efisiensi terbaik.

SCREENSHOT: Recommendation — Tanaman Sakit(sick plants)

AgriTech

DashboardFarmsCropsRekomendasiHarga PasarBelanjaForumAhlAdmin

YunaLogout

Rekomendasi Pertanian

Panduan pemupukan dan irigasi berdasarkan kondisi tanaman

Pilih Tanaman untuk Rekomendasi

PILIH FARM

Sawah Padi

PILIH TANAMAN

Padi (growing) Sakit

Padi

Stage: growing

Status: Sakit

Tanaman ini dalam kondisi sakit! Segera lakukan penanganan berikut:

Identifikasi Penyakit

Periksa gejala: bercak daun, layu, perubahan warna. Konsultasikan dengan penyuluh pertanian.

Karantina Tanaman Sakit

Pisahkan tanaman sakit dari yang sehat untuk mencegah penyebaran penyakit.

Sanitasi Lahan

Buang bagian tanaman sakit dan bakar. Bersihkan alat pertanian setelah dipakai.

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Gunakan fungisida berbahan aktif Mankozeb atau Klorotalonil untuk penyakit jamur. Ikuti dosis anjuran.

Atur Kelembaban

Pastikan drainase baik. Kelembaban berlebih mempercepat penyebaran jamur dan bakteri.

Pupuk untuk Fase Growing

JENIS PUPUK	DOSIS	WAKTU
Pupuk NPK Susulan (15-9-20) Untuk pertumbuhan vegetatif	100 kg/ha	30 HST

Irigasi untuk Fase Growing

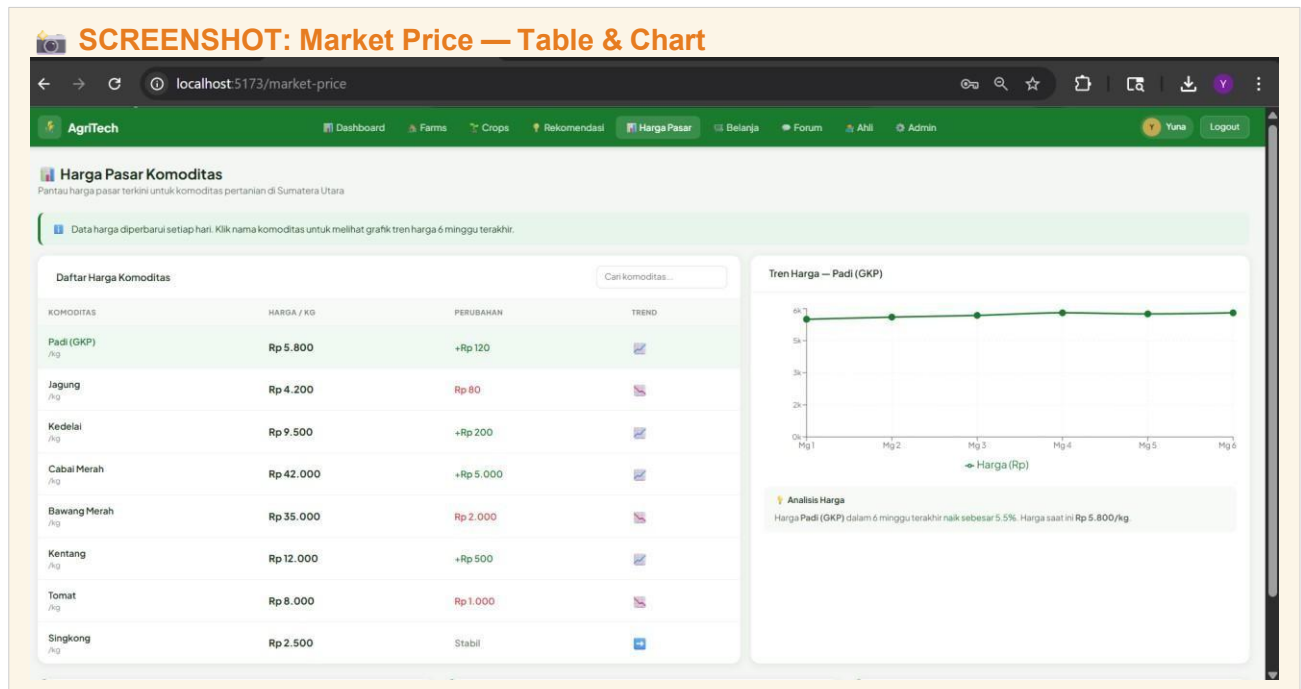
Frekuensi Penyiraman

1x sehari atau sesuai kondisi

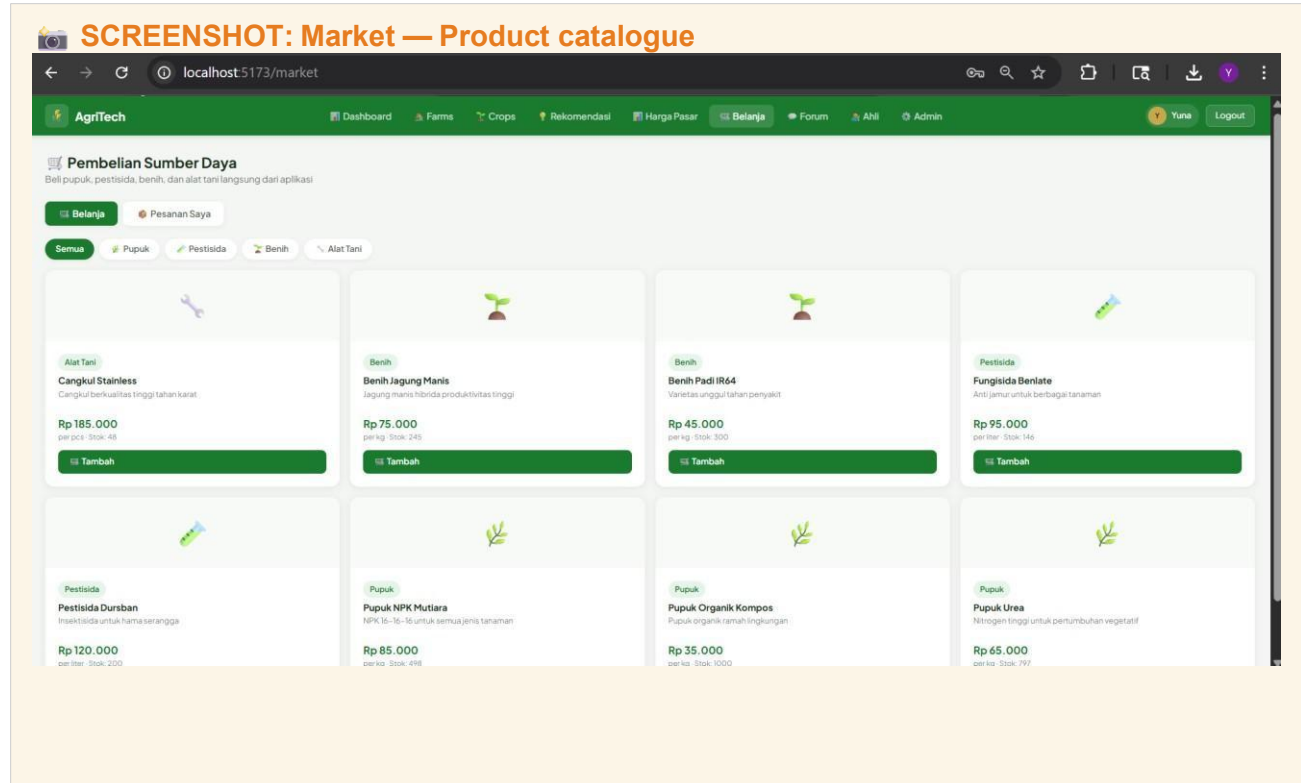
Volume Air

5-8 liter/m²/hari

➤ Market Price



➤ Resource Purchase



SCREENSHOT: Market — Shopping cart

AgriTech

DashboardFarmsCropsRekomendasiHarga PasarBelanjaForumAhlAdmin

YunaLogout

Pembelian Sumber Daya

Beli pupuk, pestisida, benih, dan alat tani langsung dari aplikasi

BelanjaPesanan Saya

Keranjang (1) — Rp 85.000

Keranjang Belanja

Pupuk NPK Mutiara

Rp 85.000 /kg

- 1 +

Total:Rp 85.000

ALAMAT PENGIRIMAN

Berastagi

Konfirmasi Pesanan

SemuaPupukPestisidaBenihAlat Tani

Pupuk

Pupuk NPK Mutiara

NPK 16-16-16 untuk semua jenis tanaman

Rp 85.000

Pupuk

Pupuk Organik Kompos

Pupuk organik ramah lingkungan

Rp 35.000

Pupuk

Pupuk Urea

Nitrogen tinggi untuk pertumbuhan vegetatif

Rp 65.000

SCREENSHOT: Market — Order history

AgriTech

DashboardFarmsCropsRekomendasiHarga PasarBelanjaForumAhlAdmin

YunaLogout

Pembelian Sumber Daya

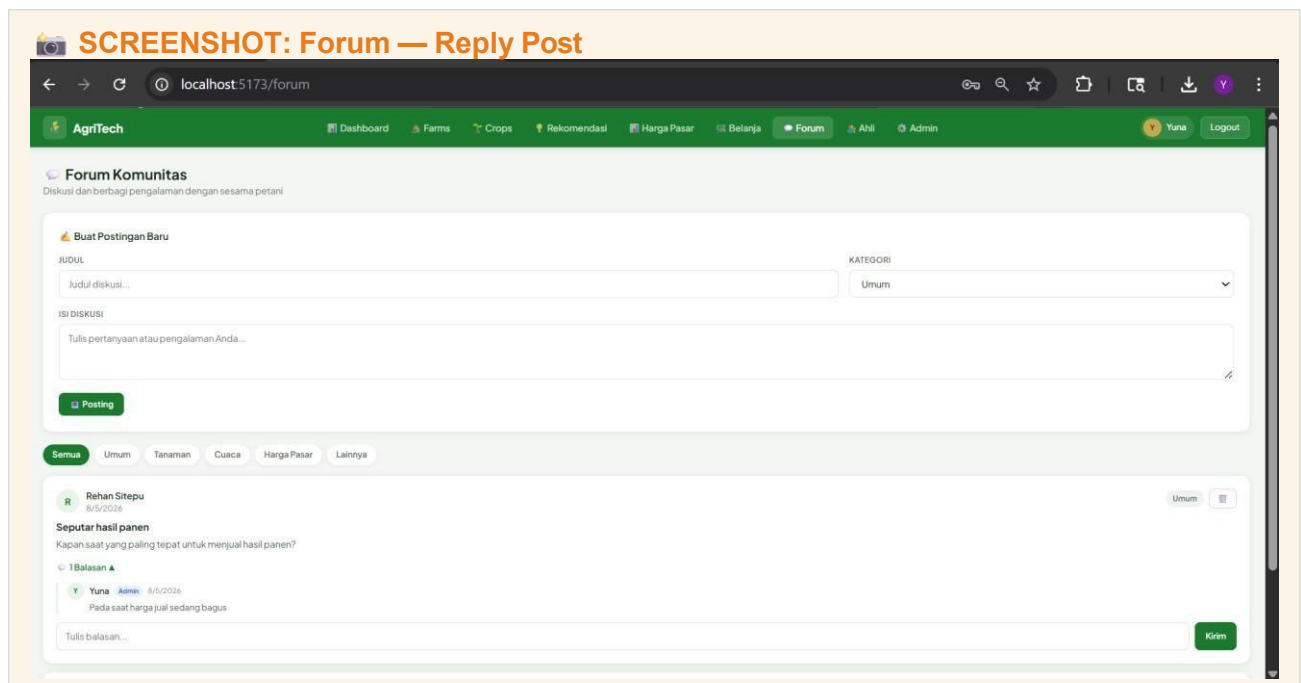
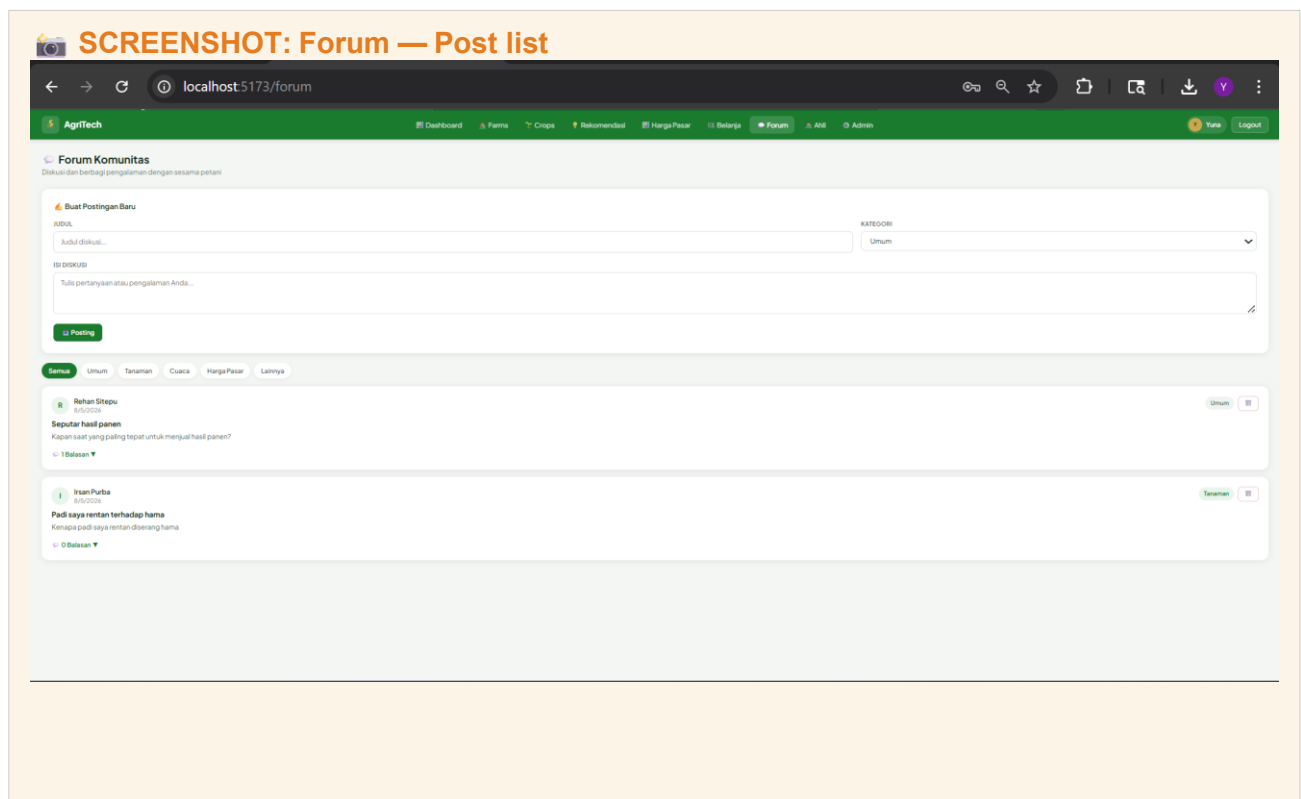
Beli pupuk, pestisida, benih, dan alat tani langsung dari aplikasi

BelanjaPesanan Saya

Riwayat Pesanan (4)

TANGGAL	PRODUK	TOTAL	ALAMAT	STATUS
11/5/2026	Pestisida Dursban(2)	Rp 240.000	Berastagi	Pending
11/5/2026	Pupuk NPK Mutiara (2)	Rp 170.000	Berastagi	Pending
11/5/2026	Benih Jagung Manis (5)	Rp 375.000	Berastagi	Pending
11/5/2026	Cangkul Stainless (2)	Rp 370.000	Berastagi	Pending

➤ Community Forum



➤ Expert advice

SCREENSHOT: Expert — Pending questions

localhost:5173/expert

AgriTech Dashboard Farms Crops Rekomendasi Harga Pasar Belanja Forum Ahli Irsan Purba Logout

Kirim Pertanyaan

Semua Belum Dijawab (1) Sudah Dijawab (2)

I Irsan Purba
Duru saja
Mengapa padi saya sering kali gagal panen?

Tanaman Pending

R Rehan Sitepu
8/5/2026
Kenapa hama pada tanaman tomat saya begitu mengerikan?

Hama & Penyakit Terjawab

JAWABAN AHLI — YUNA
Serangan hama pada tanaman tomat terasa "mengerikan" karena jenis hama yang menyerang seringkali bekerja secara cepat, bergerombol, dan merusak berbagai bagian tanaman sekaligus (daun, batang, buah). Akibatnya, tanaman bisa layu, daun keriting, buah busuk, bahkan gagal panen.

I Irsan Purba
8/5/2026
Padi saya banyak yang menguning, apa solusi yang tepat?

Tanaman Terjawab

JAWABAN AHLI — YUNA
Daun padi menguning dapat diatasi dengan mengidentifikasi penyebab utamanya: asam-asaman (perbaiki drainase), kekurangan nutrisi (pupuk Nitrogen/Kalium), atau penyakit (fungisida/bakterisida). Langkah cepat meliputi pengeringan sawah, aplikasi dolomit (jika tanah masam), dan penggunaan pupuk berimbang.

SCREENSHOT: Expert — Questions answered

localhost:5173/expert

AgriTech Dashboard Farms Crops Rekomendasi Harga Pasar Belanja Forum Ahli Admin Yuna Logout

Saran Ahli Pertanian
Tanyakan masalah pertanian Anda dan dapatkan jawaban dari ahli

Mode Admin: Anda dapat menjawab pertanyaan dari petani di bawah ini.

Semua Belum Dijawab (1) Sudah Dijawab (2)

I Irsan Purba
1 menit lalu
Mengapa padi saya sering kali gagal panen?

Tulis jawaban Anda sebagai ahli...

Jawab

Tanaman Pending

R Rehan Sitepu
8/5/2026
Kenapa hama pada tanaman tomat saya begitu mengerikan?

Hama & Penyakit Terjawab

JAWABAN AHLI — YUNA
Serangan hama pada tanaman tomat terasa "mengerikan" karena jenis hama yang menyerang seringkali bekerja secara cepat, bergerombol, dan merusak berbagai bagian tanaman sekaligus (daun, batang, buah). Akibatnya, tanaman bisa layu, daun keriting, buah busuk, bahkan gagal panen.

I Irsan Purba
8/5/2026
Padi saya banyak yang menguning, apa solusi yang tepat?

Tanaman Terjawab

JAWABAN AHLI — YUNA

➤ Admin Panel

SCREENSHOT: Admin — Overview Stats

localhost:5173/admin

AgriTech Dashboard Farms Crops Rekomendasi Harga Pasar Belanja Forum Ahli Admin Yuna Logout

Admin Panel

Kelola sistem dan pengguna AgriTech

Overview Manage Users

TOTAL FARMERS
4

TOTAL FARMS
14

TOTAL CROPS
35

CROPS SAKIT
12

Pengguna Terbaru

NAMA	EMAIL	ROLE	BERGABUNG
Juniko Ginting	juniko@gmail.com	farmer	7/5/2026
Rehan Sitepu	rehan@gmail.com	farmer	7/5/2026
Irsan Purba	irsan@gmail.com	farmer	7/5/2026
Yuna S	yuna2@gmail.com	farmer	7/5/2026

SCREENSHOT: Admin — Manage Users

localhost:5173/admin

AgriTech Dashboard Farms Crops Rekomendasi Harga Pasar Belanja Forum Ahli Admin Yuna Logout

Admin Panel

Kelola sistem dan pengguna AgriTech

Overview Manage Users

Semua Pengguna (5)

#	NAMA	EMAIL	ROLE	BERGABUNG	AKSI
1	Juniko Ginting	juniko@gmail.com	farmer	7/5/2026	Hapus
2	Rehan Sitepu	rehan@gmail.com	farmer	7/5/2026	Hapus
3	Irsan Purba	irsan@gmail.com	farmer	7/5/2026	Hapus
4	Yuna S	yuna2@gmail.com	farmer	7/5/2026	Hapus
5	Yuna	yuna1@gmail.com	admin	6/5/2026	— Anda —

BAB 4 — Backend Development

4.1 Middleware Pipeline

Express Middleware Chain

Incoming request:

- ↓ `cors()` – izinkan cross-origin dari `localhost:5173`
- ↓ `express.json()` – parse body JSON
- ↓ Router matching – cocokkan URL ke route yang tepat
- ↓ `authMiddleware` – verifikasi JWT token (jika protected route)
- ↓ `roleMiddleware` – cek role admin (jika admin-only route)
- ↓ Controller function – jalankan business logic
- ↓ Response JSON – kirim hasil ke client
- ↓ Error Handler – tangkap semua error, return 500

4.2 Contoh Implementasi Controller

cropController.js — `createCrop()`

```
exports.createCrop = async (req, res, next) => {  
  try {  
    const { farmId, name, stage, healthStatus } = req.body  
    // Validasi input  
    if (!farmId || !name) return res.status(400).json({  
      success: false, message: 'farmId and name required'  
    })  
    // Cek farm exist & ownership  
    const farm = await Farm.findById(farmId)  
    if (!farm) return res.status(404).json({ message: 'Farm not found' })  
    if (farm.userId.toString() !== req.user.id)  
      return res.status(403).json({ message: 'Unauthorized' })  
    // Buat crop baru  
    const crop = await Crop.create({ farmId, name, stage, healthStatus })  
    res.status(201).json({ success: true, data: crop })  
  } catch (error) { next(error) }  
}
```

4.3 User Stories & Incoming request

User Story	Skenario Implementasi
As a farmer, I want to monitor my crop health	Budi logs in → opens Crops → selects 'Main Field' → sees Padi IR64 status Sick → system automatically shows red alert on Dashboard → Budi clicks 'Handle Now' → taken to Recommendations page
As a farmer, I want to know which fertilizer to use	Yuna opens Recommendations → selects farm → selects crop 'Sweet Corn' stage Growing → system displays: NPK Sustainer 100 kg/ha, KCI 50 kg/ha, specific application timing
As a farmer, I want to know today's weather	Farmer opens Dashboard → Medan weather auto-detects 28°C, Humidity 78% → types 'Langkat' in search → Langkat weather 31°C appears → farmer decides to postpone spraying due to strong winds
As an admin, I want to answer farmer questions	Admin logs in → opens Expert Advice → sees pending questions from farmers → fills in answer → clicks 'Answer' → status changes to 'Answered' → farmer gets answer notification
As a farmer, I want to buy fertilizer	Farmer opens Market → filters 'Fertilizer' → adds 'NPK Mutiara' 2 kg to cart → fills address → confirms order → stock decreases automatically → order appears in history with Pending status

BAB 5 — Database Design

5.1 Schema MongoDB — Details

Users Collection:

models/User.js

```
const userSchema = new mongoose.Schema({
  name:      { type: String, required: true, trim: true },
  email:     { type: String, required: true, unique: true, lowercase: true },
  password:  { type: String, required: true }, // bcrypt hash
  role:      { type: String, enum: ['farmer','admin'], default: 'farmer' },
}, { timestamps: true }) // auto: createdAt, updatedAt
```

Farm Collection:

models/Farm.js

```
const farmSchema = new mongoose.Schema({
  userId:    { type: ObjectId, ref: 'User', required: true },
  name:      { type: String, required: true, trim: true },
  location:  { type: String, required: true, trim: true },
  size:      { type: Number, required: true }, // dalam Hektar
}, { timestamps: true })
```

Crop Collection:

models/Crop.js

```
const cropSchema = new mongoose.Schema({
  farmId:    { type: ObjectId, ref: 'Farm', required: true },
  name:      { type: String, required: true },
  stage:     { type: String, enum: ['seedling','growing','harvest'] },
  healthStatus: { type: String, enum: ['healthy','sick'] },
  plantingDate: { type: Date, default: Date.now },
}, { timestamps: true })
```

5.2 MongoDB Document Example

Contoh Document — users collection

```
{
  _id: ObjectId('64f8a1b2c3d4e5f6a7b8c9d0'),
  name: 'Yuna Yeneskia',
  email: 'yunal@gmail.com',
  password: '$2b$10$rzEkE3uBSfyqDaevGrxttOiFH3...', // bcrypt hash
  role: 'farmer',
  createdAt: ISODate('2026-05-06T12:54:44.400Z'),
  updatedAt: ISODate('2026-05-06T12:54:44.400Z')
}
```

Contoh Document — crops collection

```
{
  _id: ObjectId('65a1b2c3d4e5f6a7b8c9d0e1'),
  farmId: ObjectId('64f9b3a4c5d6e7f8a9b0c1d2'),
  name: 'Padi IR64',
  stage: 'growing',
  healthStatus: 'sick',
  plantingDate: ISODate('2026-04-15T00:00:00.000Z'),
  createdAt: ISODate('2026-04-15T08:30:00.000Z'),
  updatedAt: ISODate('2026-05-10T14:20:00.000Z')
}
```

5.3 Relationships Between Collections

Entity Relationship Diagram

users	—(1:N)→	farms	[users._id = farms.userId]
farms	—(1:N)→	crops	[farms._id = crops.farmId]
users	—(1:N)→	forums	[users._id = forums.userId]
users	—(1:N)→	orders	[users._id = orders.userId]
users	—(1:N)→	experts	[users._id = experts.userId]
users	—(1:N)→	experts	[users._id = experts.answeredBy]

Forums use EMBEDDED DOCUMENTS for replies:

```
forums.replies[] = [{ userId: ObjectId, content: String, timestamps }]
```

→ Efficient: 1 query gets post + all replies at once

BAB 6 — API Documentation

6.1 API conventions

Base URL: `http://localhost:5000/api/v1`

Format Request: Content-Type: `application/json`

Format Response: `{ success: boolean, data: any, message: string }`

Protected Route: Header Authorization: `Bearer <JWT_TOKEN>`

6.2 Authentication Endpoints

Method + Endpoint	Request Body	Response
POST /auth/register	<code>{ name, email, password }</code>	201: <code>{ success: true, message: 'Registered' }</code> 400: <code>{ message: 'Email already registered' }</code>
POST /auth/login	<code>{ email, password }</code>	200: <code>{ success: true, data: { token, user: { id, name, email, role } } }</code> 401: <code>{ message: 'Invalid credentials' }</code>

6.3 Farm & Crop Endpoints

Method + Endpoint	Access	Function & Response
GET /farms	 Protected	Get all farms owned by the logged in user Response: <code>{ data: [farms array] }</code>
POST /farms	 Protected	Create a new farm Body: <code>{ name, location, size }</code> Response 201: <code>{ data: farm }</code>
PUT /farms/:id	 Protected	Update own farm Body: <code>{ name?, location?, size? }</code> Response: <code>{ data: updatedFarm }</code>
DELETE /farms/:id	 Protected	Delete own farm Response: <code>{ message: 'Farm deleted' }</code>
GET /crops/:farmId	 Protected	Get all crops on a specific farm Response: <code>{ data: [crops array] }</code>
POST /crops	 Protected	Add new crop Body: <code>{ farmId, name, stage, healthStatus }</code> Response 201: <code>{ data: crop }</code>
PUT /crops/:id	 Protected	Update crop Body: <code>{ name?, stage?, healthStatus? }</code>
DELETE /crops/:id	 Protected	Delete crop Response: <code>{ message: 'Deleted' }</code>

6.4 Other Endpoints

Method + Endpoint	Access	Function
GET /dashboard		Statistics: totalFarms, totalCrops, healthyCrops, sickCrops
GET /weather?city=Medan		OpenWeatherMap weather data — temp, humidity, wind, pressure
GET /forum		All forum posts with replies (populated)
POST /forum		Create post — body: { title, content, category }
POST /forum/:id/reply		Reply to post — body: { content }
GET /expert		All consultation questions with status
POST /expert		Ask a question — body: { question, category }
PUT /expert/:id/answer	 Admin	Answer the question — body: { answer }
GET /products		Product catalog (auto-seed if empty)
POST /products/order		Create an order — body: { items[], address }
GET /products/my-orders		User order history
GET /admin/stats	 Admin	System statistics: users, farms, crops, sick
GET /admin/users	 Admin	List all users
PATCH /admin/users/:id/role	 Admin	Change user role — body: { role }
DELETE /admin/users/:id	 Admin	Remove the user from the system

BAB 7 — Security & Authentication

7.1 JWT Authentication Flow

Complete Authentication Flow
REGISTER:
User → POST /auth/register → validation → bcrypt.hash(password, 10)
→ User.create() → 201 response
LOGIN:
User → POST /auth/login → User.findOne({email})
→ bcrypt.compare(input, hash) → jwt.sign({id, role}, SECRET, {expiresIn:'1d'})
→ return { token, user }
→ Frontend: localStorage.setItem('token', token)
PROTECTED REQUEST:
Frontend → Axios: headers.Authorization = 'Bearer ' + token
→ authMiddleware: jwt.verify(token, SECRET) → decode {id, role}
→ req.user = {id, role} → next() → Controller
TOKEN EXPIRED:
jwt.verify() throw JsonWebTokenError → 401 Unauthorized
→ Frontend: redirect ke /login

7.2 Role-Based Access Control

Role	Access rights	No Accessible Endpoints
Public (not logged in)	There isn't any	POST /auth/register POST /auth/login
Farmer	Your own data only	GET/POST/PUT/DELETE /farms (own) GET/POST/PUT/DELETE /crops (own) GET /dashboard, /weather GET/POST /forum, /expert GET /products, POST /products/order
Admin	All + admin features	All farmer permissions + GET/DELETE /admin/users PATCH /admin/users/:id/role GET /admin/stats PUT /expert/:id/answer

7.3 Five Layers of Security

Layer	Implementation
1. Input Validation	Regex email, min password 6 char, required field check on each controller
2. Password Encryption	<code>bcrypt.hash(password, 10)</code> — 1024 iterations. No plain text is stored.
3. JWT Authentication	Each protected route verifies the Bearer token. Expired → 401.
4. Ownership Validation	<code>farm.userId === req.user.id</code> before every CRUD operation. Does not match → 403.
5. Environment Security	JWT_SECRET, MONGO_URI, API_KEY in .env — not pushed to repository.

BAB 8 — Data Analytics & Feature Implementation

8.1 AgriTech Analytical System

Analytical Type	Data source	Insights Generated
Crop Health Rate	Crop.countDocuments({healthStatus:'healthy'}) / total	Percentage of healthy plants — trigger alert if < 70%
Sick Crop Alert	Crop.countDocuments({healthStatus:'sick'}) > 0	Red alert + specific diseased plant name
Farm Statistics	Farm.countDocuments({userId})	Total farms managed by the user
Market Price Trend	Static data 6 weeks per commodity	Percentage increase/decrease in price — selling strategy
Admin System Stats	countDocuments() semua collections	Overall health of the platform ecosystem

8.2 Recommendation Engine — Detailed Logic

Planting Phase	Fertilizer Recommendations	Irrigation Recommendations
Seedling (0-14 HST)	NPK Starter 16-16-16: 50 kg/ha at planting. Urea: 25 kg/ha, 7 days after planting. Compost: 2 tons/ha before planting.	2x/day (morning & evening) 3-5 liters/m ² /day Humidity: 60-70%
Growing (15-60 HST)	Follow-up NPK 15-9-20: 100 kg/ha, 30 days after planting. KCl: 50 kg/ha, 45 days after planting. Gandasil D: 2g/liter every 2 weeks.	1x/day 5-8 liters/m ² /day Humidity: 70-80%
Harvest (>60 HST)	K2SO4: 75 kg/ha, 2 weeks before harvest Boron: 1 kg/ha, 10 days before harvest Avoid excess nitrogen	Reduce gradually by 2-3 liters/m ² /day Humidity: 40-50%

BAB 9 — Testing

9.1 Functional Testing

Test Case	Input	Expected Result — Status
Register successful	Valid field, unique email	201 Created
Register duplicate emails	Email already exists	400 'Email already registered'
Login valid	Email & password are correct	200 + JWT token
Wrong login password	Passwords do not match	401 'Invalid credentials'
Access without tokens	No Authorization header	401 'No token provided'
Farmer access /admin	role=farmer to admin route	403 'Access denied'
Add valid farm	name, location, complete size	201 Farm saved
Edit other people's farms	farmId belongs to another user	403 Unauthorized
Set crop Sick	healthStatus: 'sick'	Red alert appears on Dashboard
Health rate < 70%	More sick	Yellow alert appears
Checkout with sufficient stock	qty <= stock	201 Order created
Checkout stock is insufficient	qty > stock	400 'Insufficient stock'
Expert: farmer asks	question + category	201 status: pending
Expert: admin answer	answer from admin	200 status: answered

9.2 Integration Testing

Scenario	Results
Register → Login → Access Dashboard → Logout → Try accessing → Redirect login	Full auth flow
Add farm → Add crop → Open dashboard → Stat cards update	Data sinkron

Search weather → OpenWeatherMap → Data appears in UI	Integrated API
Set the Sick crop → Dashboard → Alert appears → Change to Healthy → Alert disappears	Real-time
Add cart → Checkout → Reduced stock → History appears	Transaction
Farmer asks → Admin answers → Status changes → Appears on the frontend	Role-based

BAB 10 — Social & Economic Impact

10.1 Quantitative Impact Projections

Indicator	Impact Projections
Reducing crop losses	Early detection crop monitoring → faster handling → estimated loss reduction of 35-40%
Fertilizer efficiency	Planting phase-based dosage → 15-25% reduction in over-application → saving Rp. 500,000-1,500,000/ha/season
Water efficiency	Condition-based irrigation guide → 20-30% reduction in water consumption
Increased selling margin	Real-time price access → sales timing optimization → +10-20% margin

10.2 Social & Environmental Impacts

- Democratization of access to information — smallholder farmers gain equal access to large agribusinesses
- Reduction of dependence on middlemen — price transparency provides a stronger bargaining position
- Knowledge transfer — community forums digitally document local wisdom
- Reduction of agrochemical pollution — precise fertilizer dosages reduce nitrogen runoff
- Water conservation — data-driven irrigation guidelines prevent overwatering

10.3 Future Improvements

1. IoT sensor integration for real-time soil condition monitoring
2. Machine learning for historical data-based crop yield prediction
3. React Native mobile application for field access
4. Push notifications for extreme weather and market price alerts
5. Banking/cooperative system integration for platform-based microcredit

BAB 11 — Conclusion

11.1 Mapping 11 Objectives to Achievements

Objective	Achievements
1. The role of technology in agriculture	Digital logging platform, weather API, data analytics
2. Full-stack MERN	React+Vite, Express+Node, MongoDB — complete
3. Responsive UI	Bootstrap 5, 10 pages, role-based protected routes
4. RESTful API	18+ endpoints, semua HTTP methods, JSON standard
5. MongoDB agricultural data	7 collections, ObjectId relations, embedded documents
6. External data integration	OpenWeatherMap API, commodity market price data
7. Crop monitoring & recommendations	Alert threshold, 3-phase rule-based engine
8. Data analytics	8. Data analytics5 types of analytics, Recharts visualization, alert system
9. Security & validation	5 layers of security, JWT, bcrypt, RBAC, ownership check
10. Scalable system	MVC, stateless JWT, modular, horizontal scaling ready
11. Social & economic impacts	Quantitative projections, social & environmental impacts are discussed

— End of AgriTech Documentation —

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